

JPA Annotations – Hibernate Annotations

- JPA annotations are used in mapping java objects to the database tables, columns etc.
- Hibernate is the most popular implement of JPA specification and provides some additional annotations.
- Annotations are present in `javax.persistence` package.

- Java annotation is a form of metadata that can be added to Java source code.
- Java annotations can be read from source files.
- It can also be embedded in and read from class files generated by the compiler.
- This allows annotations to be retained by JVM at run-time.
- JPA annotations are not part of standard JDK, so you will get it when you add any implementation framework.

1. `javax.persistence.Entity`: Specifies that the class is an entity. This annotation can be applied on Class, Interface of Enums.
2. `@Table`: It specifies the table in the database with which this entity is mapped.
3. `@Column`: Specify the column mapping using `@Column` annotation. Name attribute of this annotation is used for specifying the table's column name.
4. `@Id`: This annotation specifies the primary key of the entity.
5. `@GeneratedValue`: This annotation specifies the generation strategies for the values of primary keys.
6. `@Version`: We can control versioning or concurrency using this annotation.
7. `@OrderBy`: Sort your data using `@OrderBy` annotation.
8. `@Transient`: Every non static and non-transient property of an entity is considered persistent, unless you annotate it as `@Transient`.
9. `@Lob`: Large objects are declared with `@Lob`.
10. `@PrimaryKeyJoinColumn`
This annotation is used to associate entities sharing the same primary key.
11. `@JoinColumn`
`@JoinColumn` annotation is used for one-to-one or many-to-one associations when foreign key is held by one of the entities.
12. `@JoinTable`: `@JoinTable` and `mappedBy` should be used for entities linked through an association table.

- 13. @DiscriminatorColumn:** As the name suggests this column is the discriminator and this annotation specifies the discriminator column for the SINGLE_TABLE and JOINED Inheritance mapping strategies.
- 14. @OneToMany** – this annotation is applied on a collection property to manage one to many relationship. Use mappedBy attribute for bidirectional associations.
For a unidirectional association with join column approach , we need to use @JoinColumn annotation in conjunction with @OneToMany annotation.
- 15. @ManyToOne-** this annotation is used to manage one to many relationship with @JoinColumn to specify the join column in entities.
- 16. @ManyToMany** – this annotation is used to manage many to many relationship with @JoinTable annotation to specify the join table.
- 17. @OneToOne-** this annotation is used to manage one to one relationship with @JoinColumn annotation to specify foreign key- primary key relationship